

MATHIS SEIDL

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EDUCATION

BOSTON UNIVERSITY

Boston, MA

B.S. in Mechanical Engineering, graduating in May 2027

2023 – 2027

- **GPA:** 3.55/4.00
- **Concentration:** Robotics (Machine Design, Automation & Manufacturing Methods, Controls, Dynamics, Robotics 1)
- **Projects:** Inventory Inspection Robot, Mailroom Robotic Arm, Connect-4 Game, Skittles Color Sorter

WORK EXPERIENCE

AMAZON

Erfurt, Germany

Operations Engineer Intern, ERF1 (Amazon Erfurt site) – RME (Reliability Maintenance Eng.) Dept.

Summer 2026

- Coordinated the rollout of Amazon's Vulcan Robot for ERF1 RME; navigated German safety/compliance requirements; built Vulcan's training program, certifying 37 technicians; POC between Amazon Robotics and RME
- Coded and launched RME's knowledge hub website in HTML, adopted by 70+ technicians, engineers, and managers
- Designed trackers used daily by all 15 Sr. Technicians to log project progress and hand off open tasks between shifts
- Used SQL to find the root cause of 200 missed shipments: a backed-up conveyor, not a labeling machine fault
- Performed 50+ Permit to Work reviews for contractors on high-risk operations, with zero safety incidents

SIEMENS

Munich, Germany

Automation Engineer Intern, R&D Dept – Industrial Metaverse Lab

Summer 2025

- Programmed an S7-1500 PLC in ladder logic to control a robotic cell – 3 arms, 2 conveyors, 4 proximity sensors
- Built a 1:1 digital twin of the robotic cell in CAD & Process Simulate, synced live to the S7-1500 PLC via OPC UA
- Defined kinematic chains from scratch for the 3 robotic arms and 2 grippers, matching their real-world motion
- Connected the cell to the Siemens AI Copilot, enabling voice commands

PROJECTS (find more inspiring projects on my [portfolio](#))

Autonomous Inventory Inspection Robot – Senior Capstone Project

Fall 2026 – Present

- Building a mobile manipulator that autonomously navigates, opens drawers, and flags missing or misplaced tools
- Designing the full robot in CAD and manufacturing it with 3D printing and CNC machining; using 2D LiDAR and SLAM to map the space and Nav2 to drive to each drawer on ROS 2

Autonomous Mailroom Robotic Arm

Spring 2026

- Built a 6 DOF static manipulator with gripper and force feedback to autonomously sort 5, 7, 9cm packages by size
- Designed the full arm in CAD, modeled kinematics in MATLAB, and programmed an ESP32 in C++ for control

Autonomous Connect-4 Game

Spring 2026

- Engineered a 3 axis belt-driven gantry with 4 stepper motors, reaching 85% chip accuracy in 10 to 20 second cycles
- Designed the full gantry in CAD, manufactured it, and wrote G-code in MATLAB to control the MKS board

TECHNICAL SKILLS

- **Programming:** Python, MATLAB, C++, G-code, SQL, HTML, PLC Ladder Logic
- **Simulation:** CAD (SolidWorks, Fusion 360), FEA (Abaqus), CFD (COMSOL), TIA Portal, Siemens Tecnomatix
- **Manufacturing:** CNC, Machining, CAM, UR PolyScope, 3D Printing, GD&T
- **Program Mgmt:** End-to-End Program Ownership, Cross-Functional Collaboration, Safety & Compliance

ADDITIONAL INFORMATION

- **Certificates:** AI Infrastructure & Operations – NVIDIA
- **Volunteering:** MIT NCAA Division III Men's Tennis Volunteer Assistant Coach, BU Resident Assistant
- **Interests:** Tennis (World Rank #976 in U18, 11.96 UTR), Skiing, Running
- **Languages:** English (native), German (native), French (proficient)